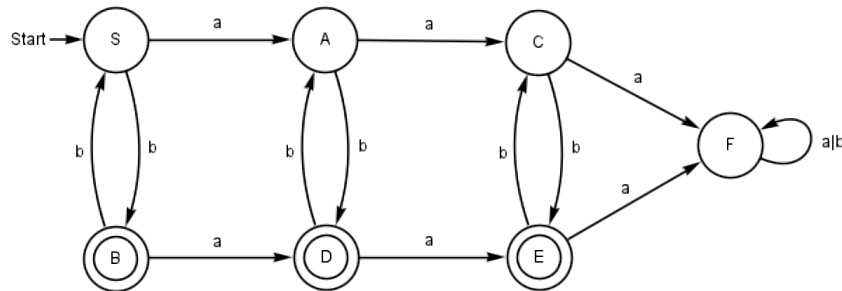


Fundamentals of theory of computation - Sample test

Each task is worth 10 points.

- What is the language generated by the following grammar?
 $G_1 = (\{S, A, B, C, D, E, F\}, \{a, b\}, S, \{S \rightarrow aA|bB, A \rightarrow aC|bD, B \rightarrow aD|bS|\varepsilon, C \rightarrow aF|bE, D \rightarrow aE|bA|\varepsilon, E \rightarrow aF|bC|\varepsilon, F \rightarrow aF|bF\})$
Solution: We can see that the grammar is in 3NF, so we may construct the following NFA (actually it's a DFA) from G_1 :



A DFA accepting $L(G_1)$

From here we can see, that the words in $L(G_1)$ must contain an even number of b 's and at most 2 a 's. So

$$L(G_1) = \{u \in \{a, b\}^* : |u|_a \leq 2 \wedge |u|_b \equiv 0 \pmod{2}\}.$$

- Give a regular (type 3) grammar which generates the following language: $L_2 = \{u \in \{0, 1\}^* \mid u \text{ is a binary number divisible by } 3\}$
 $(L_2 = \{0, 11, 110, 1001, 1100, 1111, 10010, \dots\})$
Solution: In most cases when we want to generate a regular language it's best to give a grammar in 3NF so the derivation of a terminal word

is done from letter to letter. Then we just need to take into consideration what happens when we concatenate certain letters to our words. In the case of L_2 we have to check how the remainders (after dividing by 3) change:

n	$n \bmod 3$	$n0$	$n0 \bmod 3$	$n1$	$n1 \bmod 3$
1	1	10	2	11	0
10	2	100	1	101	2
11	0	110	0	111	1
...	...	...	...	...	...
u	0	$u0$	0	$u1$	1
u	1	$u0$	2	$u1$	0
u	2	$u0$	1	$u1$	2

We can see, that if we write a 0 after our word/number, then the remainder doubles (by modulo 3), and if we write a 1, then the remainder doubles and increases by 1 (by modulo 3). So we can create a grammar in 3NF, where the nonterminal symbols represent the remainder classes 0,1 and 2:

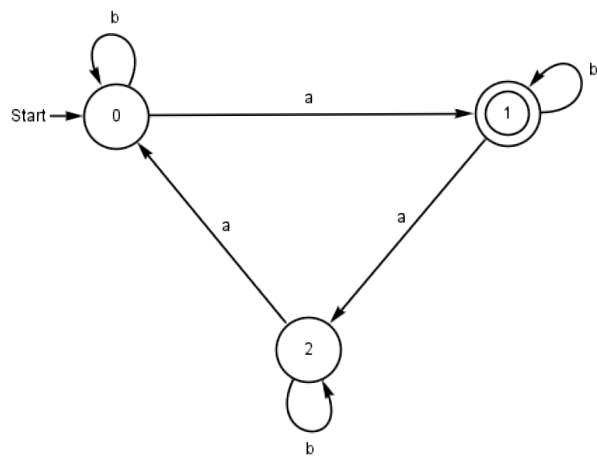
$$G_2 = (\{S, Z, O, T, E\}, \{0, 1\}, S, \{S \rightarrow 0E | 1O, Z \rightarrow 0Z | 1O | \varepsilon, \\ O \rightarrow 0T | 1Z, T \rightarrow 0O | 1T, E \rightarrow \varepsilon\})$$

(We need $S \rightarrow 0E$ and $E \rightarrow \varepsilon$ because 0 is the only number divisible by 3 which starts with 0.)

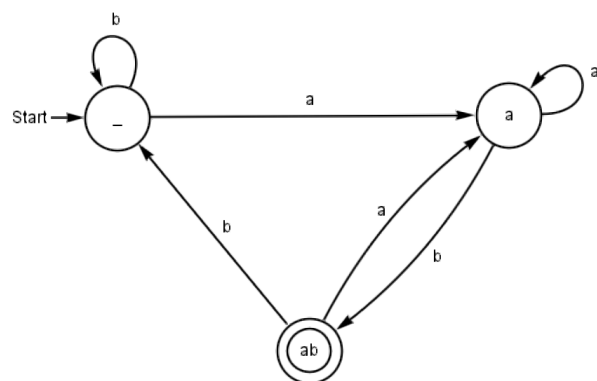
3. Give a deterministic finite automaton (DFA) which accepts the following language:

$$L_3 = \{u \in \{a, b\}^* : |u|_a \equiv 1 \pmod{3} \text{ and } u \text{ ends with 'ab'}\}$$

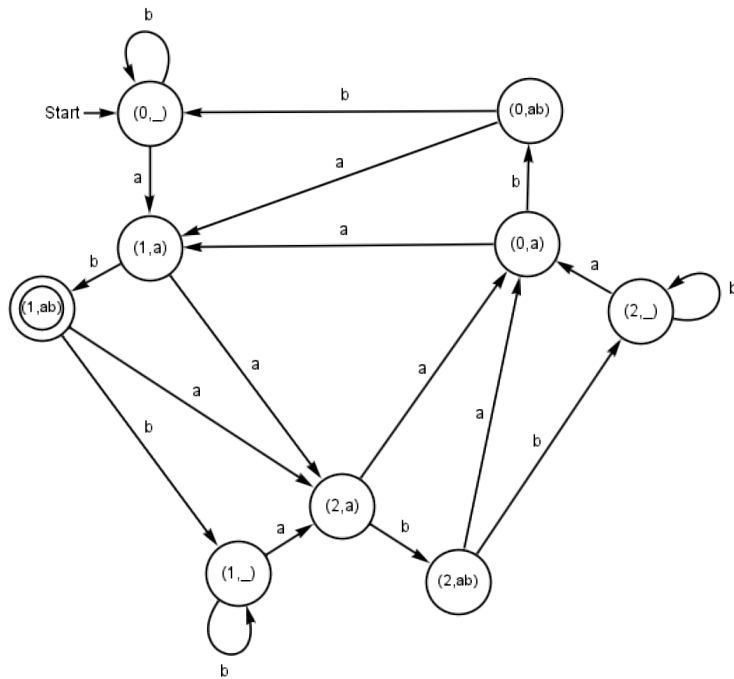
Solution: We may first construct two simpler automata accepting languages $L_{3.1} = \{u \in \{a, b\}^* : |u|_a \equiv 1 \pmod{3}\}$ and $L_{3.2} = \{u \in \{a, b\}^* : u \text{ ends with 'ab'}\}$, then create their crossproduct:



A DFA accepting $L_{3.1}$

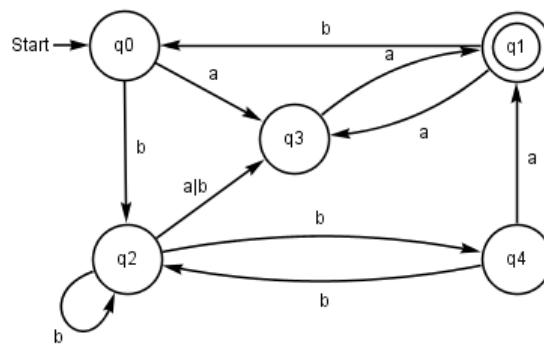


A DFA accepting $L_{3.2}$



A DFA accepting L_3

4. Give a DFA equivalent to the following non-deterministic finite automaton:



Make it deterministic.

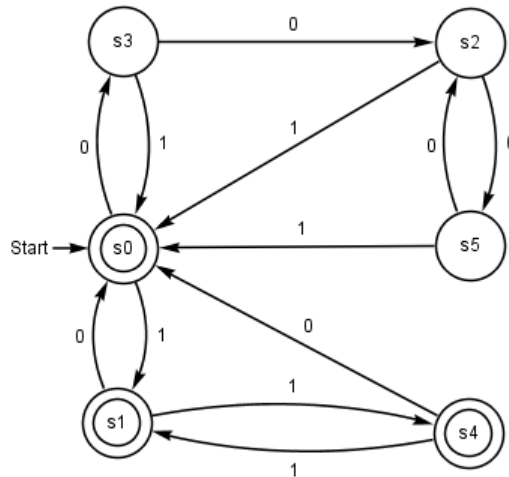
Solution: Let's give the automaton with its transition table:

δ	a	b
$\rightarrow$	q_0	q_3
$\leftarrow$	q_1	q_3
	q_2	q_3
	q_3	q_1
	q_4	q_1

In the constructed DFA the states will represent sets of states of the NFA: the initial state (of the DFA) will be the set of initial states (of the NFA); the set of accepting states will be those states, which represent a set of states (of the NFA) which include at least one accepting state; the transition function is created by taking the unions of corresponding rows in the transition table of the NFA. The transition table of the DFA will be the following:

δ'	a	b
$\rightarrow$	$\{q_0\}$	$\{q_3\}$
	$\{q_3\}$	$\{q_1\}$
	$\{q_2\}$	$\{q_3\}$
$\leftarrow$	$\{q_1\}$	$\{q_3\}$
	$\{q_2, q_3, q_4\}$	$\{q_1, q_3\}$
$\leftarrow$	$\{q_1, q_3\}$	$\{q_1, q_3\}$

5. Minimize the following DFA:



Minimize it.

Solution: First we must check if there are unreachable states in the automaton, those must be removed (in this case every state is reachable). After that we must contract equivalent states. To do this, we must determine the *i-indistinguishability* classes:

- $\overset{0}{\sim}$: $\{q_0, q_1, q_4\}, \{q_2, q_3, q_5\}$ (accepting and non-accepting states)
- $\overset{1}{\sim}$: $\{q_0\}, \{q_1, q_4\}, \{q_2, q_3, q_5\}$ (q_0 is separated from q_1 and q_4 because for the input symbol 0 we get to a non-accepting state from it)
- $\overset{2}{\sim}$: $\{q_0\}, \{q_1, q_4\}, \{q_2, q_3, q_5\}$ (no more separation can be made, the states in the same set lead to the same sets for the same input symbols)

In the minimal DFA the states represent these equivalency classes:

	δ'	0	1
$\xrightarrow{\quad}$	q_0	q_{235}	q_{14}
	q_{235}	q_{235}	q_0
$\leftarrow$	q_{14}	q_0	q_{14}